

REFERME INVESTMENTS

System Requirements Document (SRD)

Logical System & Data Requirements for the Power BI Investment Analytics Solution

Document Control	Details
Document ID	RMI-SRD-001
Project	Investment Analytics & Decision-Support Framework
Project Period	February - June 2026
Version	1.0
Status	Portfolio / case-study baseline
Methodology control	Scoring weights, thresholds and technical implementation details require stakeholder approval before production use.

1. Purpose

Define the logical system requirements needed to implement the approved business and functional requirements. The project source does not establish a production deployment architecture, infrastructure sizing, authentication product, SLA or refresh frequency; those items are therefore identified as TBD rather than invented.

2. Logical Solution Context

Business / financial / operational / customer data -> validation and transformation -> KPI calculation -> scoring layer -> semantic data model -> Power BI decision pages -> portfolio monitoring and risk alerts.

3. System Components

Component	Responsibility	Status
Source data layer	Provide business, financial, operations, customer, time, scoring and data-quality inputs	Required; physical systems TBD
Data validation / transformation layer	Standardize types, validate required inputs and prepare KPI-ready data	Required
KPI calculation layer	Calculate 28 KPI values using approved definitions	Required
Scoring layer	Normalize KPI results and calculate five dimension scores / overall Investment Score	Required; thresholds/weights configurable
Semantic model	Relate business, date, KPI, evaluation, targets and status data	Recommended logical design
Power BI report layer	Provide five decision-oriented pages and drill-down	Required
Configuration / assumptions layer	Store approved weights, thresholds and methodology parameters	Required for controlled scoring

Monitoring / alert logic	Evaluate actual-vs-target and approved risk thresholds	Required for portfolio monitoring
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4. Logical Data Model

Logical Object	Grain / Purpose	Representative Fields
DimBusiness	One row per business	BusinessID, BusinessType, evaluation attributes
DimDate	One row per date	Date, month, quarter, year
DimKPI	One row per KPI	KPIKey, KPIName, Dimension, unit, direction
FactEvaluation	Business evaluation snapshot	Dimension scores, Investment Score, recommendation, data-quality status
FactBusinessPeriod	Business + reporting period	Raw/derived financial, operations and customer measures
FactTargets	Business + reporting period	Approved target values
FactKPIStatus	Business + period + KPI	KPI value, normalized score, status, threshold context
Assumptions / Configuration	One row per parameter	Weights, thresholds and approved methodology settings

5. System Functional Requirements

SR ID	System Requirement	Source Alignment
SR-F-01	System shall ingest or receive investment-target master and evaluation data required by FR-01.	BR/FR source-supported
SR-F-02	System shall validate required data before KPI/scoring processing.	FR-02 / FR-12
SR-F-03	System shall calculate applicable KPI values from validated inputs.	FR-02
SR-F-04	System shall support configurable normalization thresholds / benchmarks.	FR-03; approval required
SR-F-05	System shall calculate five dimension scores using approved weights.	FR-04
SR-F-06	System shall calculate the overall 0-100 Investment Score using approved dimension weights.	FR-05
SR-F-07	System shall support approved recommendation logic and risk overrides.	BR-05 / FR-05
SR-F-08	System shall provide five Power BI report pages defined in the FRD.	FR-06 to FR-10
SR-F-09	System shall support target-level filtering and drill-down.	FR-11
SR-F-10	System shall expose data-quality status with the evaluation.	FR-12
SR-F-11	System shall support historical actual-vs-target,	FR-09

	KPI, score and risk trend analysis when dated data is available.	
SR-F-12	System shall surface approved threshold breaches for investigation.	FR-10

6. Data Validation Requirements

SR-DQ ID	Requirement
SR-DQ-01	Business identifier shall be present and valid for every evaluation record.
SR-DQ-02	Period-dependent records shall contain a valid evaluation/reporting period.
SR-DQ-03	Duplicate business-period records shall be identified and resolved before scoring.
SR-DQ-04	Non-calculable KPI conditions, including zero/invalid denominators, shall be handled explicitly.
SR-DQ-05	Missing/invalid/stale required inputs shall contribute to visible data-quality status.
SR-DQ-06	Scoring configuration shall be validated before use; approved weights must reconcile to the intended total.
SR-DQ-07	Historical calculations shall use period-aligned data and shall not infer missing history.

7. Reporting & Interaction Requirements

SR-UI ID	Requirement
SR-UI-01	Overview shall present portfolio-level pipeline and score/recommendation summary.
SR-UI-02	Investment Evaluation shall support one selected target and expose score drivers.
SR-UI-03	Business Profile shall show selected-target financial, customer, operational, growth and risk information.
SR-UI-04	Portfolio Monitoring shall show actual-vs-target and historical score/risk changes when source history exists.
SR-UI-05	Risk & Alerts shall identify affected businesses and risk indicators.
SR-UI-06	Filters/drill-through shall preserve a consistent business context across related visuals.
SR-UI-07	Financial values shall use consistent units/currency as defined by the approved dataset/reporting standard.

8. Non-Functional Requirements

The source does not provide measurable production NFR targets. The following are requirement categories to be baselined with technical and business owners; no numeric SLA is assumed.

NFR ID	Category	Requirement / TBD
NFR-01	Performance	Dashboard response-time target: TBD after dataset size and deployment environment are known.
NFR-02	Availability	Report/service availability target: TBD.
NFR-03	Refresh	Data refresh frequency and cut-off schedule: TBD by business/data owners.
NFR-04	Security	Access shall be restricted to authorized users; authentication/authorization mechanism: TBD.
NFR-05	Confidentiality	Sensitive financial/customer data shall follow applicable organizational access and handling controls; specific classification: TBD.
NFR-06	Auditability	Scoring configuration, requirement changes and approved methodology should be version-controlled / traceable.
NFR-07	Usability	Users shall be able to identify score, recommendation, underlying drivers and risk/data-quality limitations without ambiguous interpretation.
NFR-08	Maintainability	KPI definitions, weights and thresholds should be configurable/documented to support approved changes without redesigning the full report.

NFR-09	Data Quality	Validation failures shall be visible rather than silently converted into valid-looking scores.
NFR-10	Compatibility	Supported Power BI/Desktop/Service environment and browser/device requirements: TBD.

9. Interfaces / Integrations

Interface	Direction	Requirement	Status
Business/financial source -> analytics layer	Inbound	Provide fields defined in the data requirements / STTM	Physical source TBD
Analytics/scoring layer -> semantic model	Internal	Persist or expose KPI, dimension, score, recommendation and quality results	Logical requirement
Semantic model -> Power BI	Internal	Expose measures/dimensions required by five report pages	Required
Targets/configuration -> monitoring/scoring	Inbound	Provide approved targets, weights and thresholds	Ownership/process TBD

10. Security & Access Considerations

Role/access model is not established by the source and must be confirmed before production deployment.

Investment-target financial and customer information should be treated according to the organization's approved data-classification policy.

If row-level security is required, business rules and user-to-portfolio mappings must be documented and tested.

Exports/sharing permissions must be aligned to approved access controls.

11. System Test / Acceptance Mapping

Test	System Capability
UT-01	KPI calculation
UT-02	Dimension scoring
UT-03	Investment scoring
UT-04	Recommendation
UT-05	Drill-down
UT-06	Portfolio trend
UT-07	Risk alert
UT-08	Business profile
UT-09	Data quality

12. Open Technical Decisions

Decision	Status
Physical source systems / file interfaces	TBD
Production Power BI workspace / deployment architecture	TBD
Refresh frequency / orchestration	TBD
Authentication / authorization / RLS design	TBD
Data retention / archival	TBD
Monitoring / operational support ownership	TBD
Final benchmark and scoring threshold storage	Requires business approval